Lab4

Camera Calibration (50%)

Warping Practice (50%)

本次Lab開始，須於當天完成Demo

1. Camera Calibration (50%)

- Get image from webcam

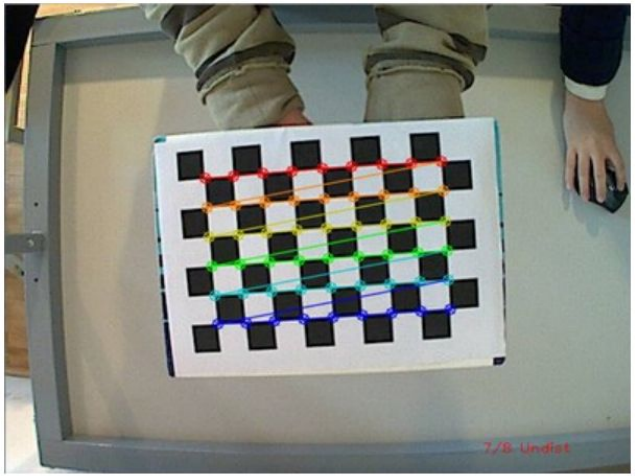
```python
import cv2

cap = cv2.VideoCapture(1)

while True:
    ret, frame = cap.read()  # ret is True if read() succeeded
    cv2.imshow("frame", frame)
    cv2.waitKey(33)
```

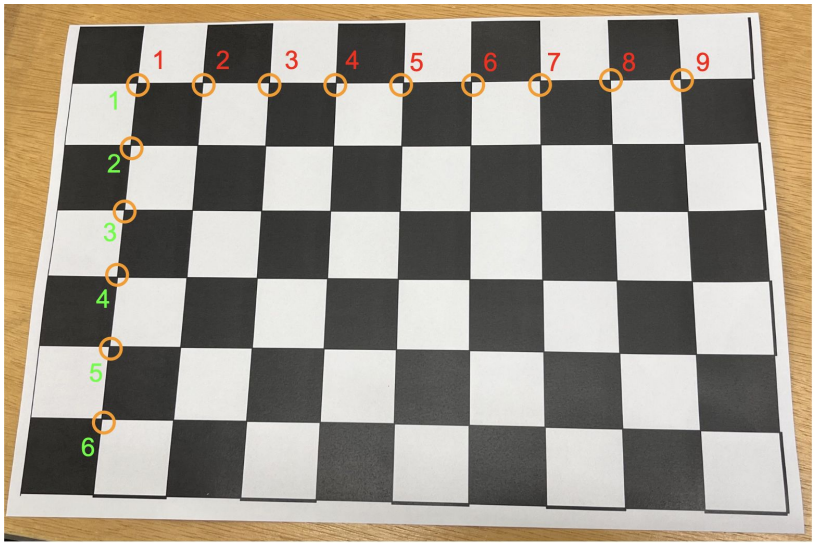
1. Camera Calibration (50%)

- 假設好棋盤格的 object point，棋盤格位於同一平面，因此通常令 Z=0
- 利用 webcam 讀取即時影像，將影像轉成灰階
- 拍攝不同角度、距離與位置的棋盤格，若有偵測到則儲存該影像中棋盤格的 image point
- 當儲存影像多於四張時，開始計算參數
- 得到參數並儲存於 xml 檔



1. Camera Calibration (50%)

- 假設棋盤格的 object point (z = 0)
- 準備 object points (0,0,0), (1,0,0), (2,0,0) ....,(8,5,0)



1. Camera Calibration (50%)

- ret, corner = cv2.findChessboardCorners(image, patternSize, None)
  - patternSize: Number of inner corners per a chessboard row and column
  - patternSize = cvSize(points_per_row,points_per_colum) = cvSize(columns,rows)
  - ret == True: chessboard detected
- cv2.cornerSubPix(image, corners, winSize, zeroZone, criteria)
  - image: Input image
  - corners: Initial coordinates of the input corners and refined coordinates provided for output.
  - winSize: (11, 11)
  - zeroZone: (-1,-1)
  - criteria = cv2.TERM_CRITERIA_EPS + cv2.TERM_CRITERIA_MAX_ITER, 30, 0.1

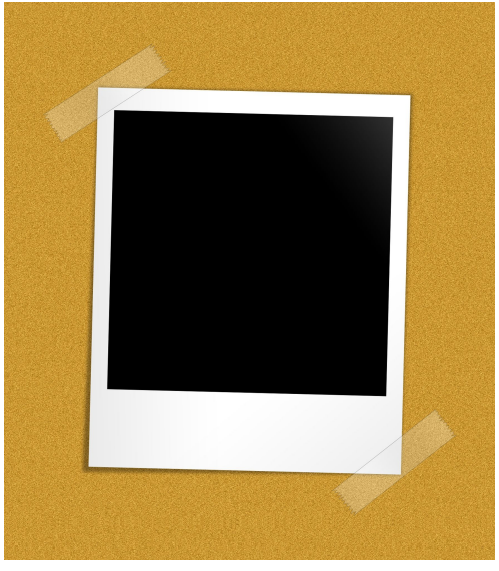
1. Camera Calibration (50%)

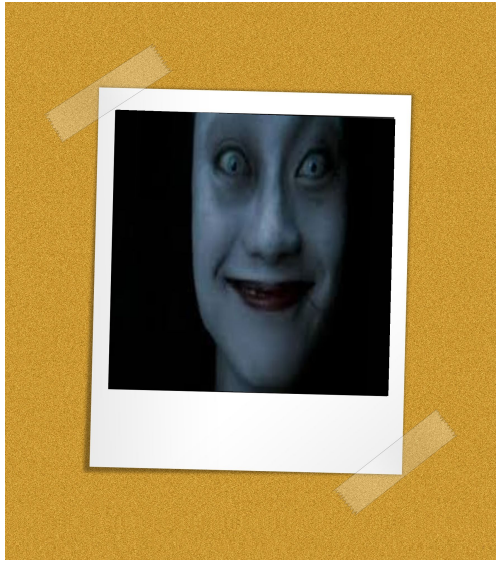
- ret, cameraMatrix, distCoeffs, rvecs, tvecs = cv2.calibrateCamera(objectPoints, imagePoints, imageSize, None)
  - cameraMatrix: Output 3x3 floating-point camera matrix
  - distCoeffs: Output vector of distortion coefficients
  - rvecs, tvecs: rotation and translation matrix
  - 有多少組 imagepoint 就要有多少組 objectpoint
- f = cv2.FileStorage(filename, cv2.FILE_STORAGE_WRITE)
  - f.write("intrinsic", mtx) //cameraMatrix
  - f.write("distortion", dist) //distCoeffs
  - f.release()

2. Warping Practice (50%)

- 透視變換 (Perspective Transform)
  - 把成像投影到一個新的視平面(Viewing Plane)
  - 也稱作投影映射 (Projective Mapping)

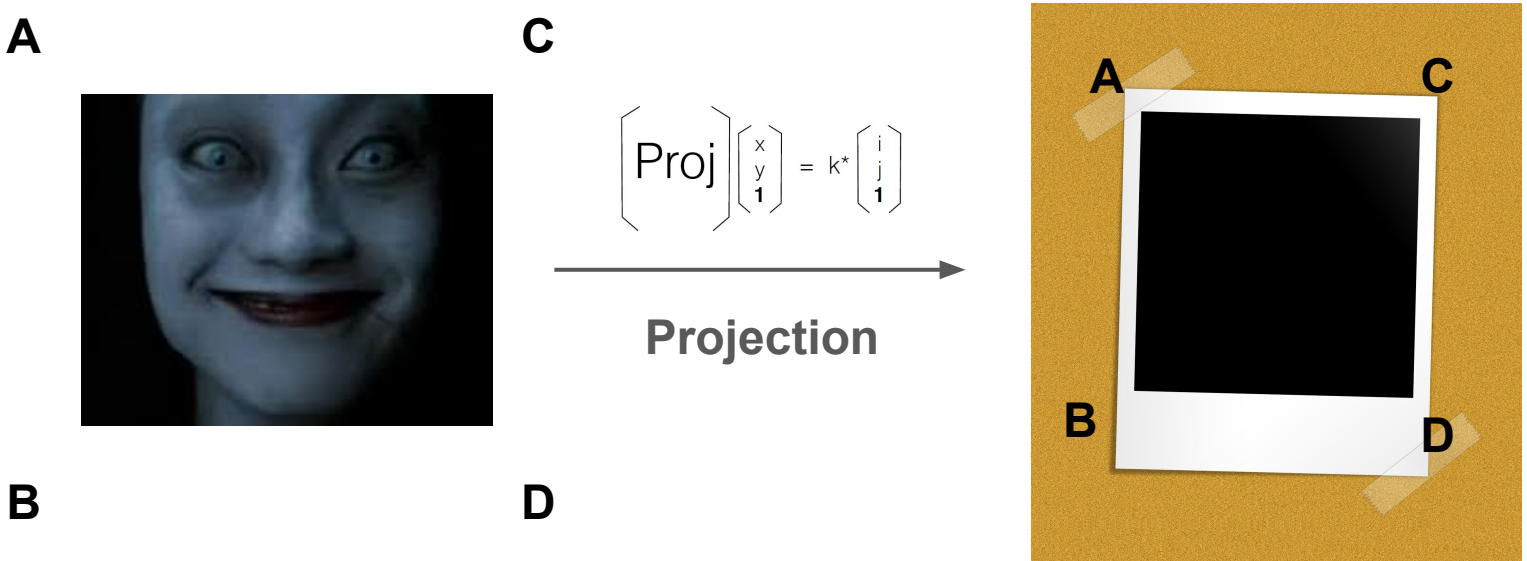






2. Warping Practice (50%)

- **cv2.getPerspectiveTransform(cap_corner, img_corner)**
  - cap_corner, img_corner 為四個點的陣列，順序需要兩兩相對
  - 返回一個 3x3 的 matrix



2. Warping Practice (50%)

只能import cv2, numpy
不可使用以下 function
cv2.warpPerspective(), cv2.remap(), cv2.resize()

- 把 webcam 得到的**即時影像 (非靜止畫面)** 同步 warp 到看板上
  - 先得到圖中對應的四個點
  - 利用 **cv2.getPerspectiveTransform** 得到轉換關係
  - 透過先前作業實作的 bilinear interpolation 適當的把圖填上








SUPER DRY
Asahi
ピップ
雪印メグミルク
道頓堀のおいしいものと
恵ガセリ菌SP株ヨーグルト
ガセリ菌SP株が
内臓脂肪を減らす
Glico
おいしさと健康
グリコ
日商エステ
かぜに
改源
CHIN
天気予報
LOTTERIA
Toilets

2. Warping Practice (50%)

不能使用 cv2.warpPerspective()

- 把 webcam 得到的**即時影像** warp 到看板上
  - 先得到兩張圖中對應的四個點
  - 利用 **cv2.getPerspectiveTransform** 得到轉換關係
  - 透過 bilinear interpolation 適當的把圖填上





